PETERSON ELEMENTS AND THE STEENROD ACTION ON THE DICKSON–MÙI ALGEBRAS

ABSTRACT. We describe the algorithm used in `yacop/modules/dickson.py` to compute the action of the mod p Steenrod algebra on the Dickson algebras D_k (and on their Mùi extensions). The algorithm rests on a family of invariants, the *Peterson elements* $\widetilde{\omega}_n$, and on a closed formula $P^R\widetilde{\omega}_n = a(R,n)\,\widetilde{\omega}_{n+d(R)}$ for the action of a Milnor basis element on them. We define the Peterson elements, give an explicit monomial formula for them which generalises a formula of Hung and Peterson, and prove that the Dickson invariants are Peterson elements. We extract the action formula from the code. For odd p it holds with the normalisation $\widetilde{\omega}_n = (-1)^n\omega_n$, not with the one stated in the code's documentation. We prove the formula for all Peterson elements, all k and all Milnor basis elements. In particular the algorithm is correct. We also prove the code's formula for the conjugate action. Finally we explain the auxiliary class `DicksonDualSteenrodAlgebra`.

1. INTRODUCTION

The program `yacop` computes the action of the Steenrod algebra $\mathcal{A}$ on the Dickson algebras $D_k = H^*(BV_k;\mathbb{F}_p)^{\mathrm{GL}_k}$, where $V_k = \mathbb{F}_p^k$. Here D_k is viewed as an algebra of invariants in the polynomial part of $H^*(BV_k)$. Since D_k is a polynomial algebra, the Cartan formula reduces the problem to the action on the generators. The code handles the generators by identifying each of them with a *Peterson element* and then using a closed formula for the Milnor basis action on Peterson elements. This note explains that formula and how much of it can be proved.

Summary of results. Let $R = (r_1, r_2, \dots)$ index the Milnor basis element P^R (written $Sq(R)$ when $p = 2$). Put

$$|R| = \sum_j r_j, \qquad N(R) = \sum_j p^j r_j, \qquad d(R) = \sum_j r_j\,\frac{p^j-1}{p-1},$$

so that $(p-1)d(R) = N(R) - |R|$. For $n \geq 0$ define the coefficient

$$a(R,n) \;=\; \binom{(p-1)(n+d(R))}{N(R)}\binom{N(R)}{pr_1,\, p^2r_2,\, p^3r_3,\, \dots} \in \mathbb{F}_p, \tag{1}$$

where the second factor is a multinomial coefficient. The code (functions `__action_exponents` and `__action_exponents2` in `dickson.py`) uses

$$P^R\,\widetilde{\omega}_n \;=\; a(R,n)\,\widetilde{\omega}_{n+d(R)}. \tag{$*$}$$

- Section 3 defines Peterson elements and proves the monomial formula and the identification of the Dickson invariants (Proposition 3.2, Theorem 3.4).
- Section 4 describes the algorithm. The code's decomposition of Peterson elements into Dickson monomials is a Lagrange inversion formula (Proposition 4.2).
- Section 5 proves ($*$) at the Dickson generators (Theorem 5.9), which shows that the algorithm is correct (Corollary 5.10). It then proves ($*$) in all degrees (Theorem 5.11), and the formula used by the code for the conjugate action (Theorem 5.14). Table 1 lists independent computer checks.

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- Section 6 explains how the code obtains the action on all of D_k (and on the Mùi generators), and Section 7 explains `DicksonDualSteenrodAlgebra`.

Remark 1.1 (A sign). The code documents its Dickson generators as the Peterson elements $\omega_n = (x_1 \cdots x_k)\,\chi(P^n)\big((x_1 \cdots x_k)^{-1}\big)$. For odd p, formula $(*)$ is *false* for the ω_n; for example $P^1\omega_1 = -\omega_2$ in $H^*(B\mathbb{Z}/3)$, while $(*)$ predicts $+\omega_2$. Formula $(*)$ is correct for $\widetilde{\omega}_n = (-1)^n\omega_n$ (Theorem 5.11, together with Corollary 5.10 for the comparison of $\widetilde{\omega}_n$ and Ω_n). Both the decomposition formula and the Cartan formula are insensitive to this sign change: n is additive under products, so a Dickson monomial in the $\widetilde{\omega}$'s picks up the same sign $(-1)^n$ as $\widetilde{\omega}_n$ itself. So the decomposition formula is the same for ω and $\widetilde{\omega}$, and the code's generators are consistent with $\widetilde{\omega}_{n_j}$. With that reading the module computed by the code is the correct one. Only the identification "generator $= \omega_n$" in the documentation should read "generator $= \widetilde{\omega}_n$". For $p = 2$ there is no difference.

2. Recollections

2.1. **Steenrod operations.** Write $H = \mathbb{F}_p[x_1, \dots, x_k]$, with $|x_i| = 1$ for $p = 2$ and $|x_i| = 2$ for p odd. This is the polynomial part of $H^*(BV_k)$. (For odd p, $H^*(BV_k) = E(y_1, \dots, y_k) \otimes H$ with $|y_i| = 1$, and the Bockstein maps $y_i \mapsto x_i$.) We use the Milnor basis $\{P^R\}$ of the subalgebra of $\mathcal{A}$ generated by the reduced powers. Let $t = (t_1, t_2, \dots)$ be indeterminates and put $t_0 = 1$. The *total Milnor operation*

$$\Lambda_t(y) = \sum_R t^R P^R(y), \qquad t^R = \prod_j t_j^{r_j},$$

is a ring homomorphism $H \to H[t]$, extended $\mathbb{F}_p[t]$-linearly. It is determined by

$$\Lambda_t(x) = T(x) := \sum_{j \geq 0} t_j\, x^{p^j} \qquad (x \in H^{|x_1|}).$$

Since the elements Λ_t are group-like, the conjugate operation $\Lambda'_t(y) = \sum_R t^R \chi(P^R)(y)$ is the inverse of Λ_t. On a degree-one (resp. degree-two) class it is $\Lambda'_t(x) = T^{-1}(x)$, where T^{-1} is the compositional inverse of the additive series T. (Indeed $\Lambda_t\big(\sum_j c_j x^{p^j}\big) = \sum_j c_j T(x)^{p^j}$ for scalars $c_j \in \mathbb{F}_p[t]$.)

2.2. **Dickson invariants.** Let $V \subset H$ be the $\mathbb{F}_p$-span of $x_1, \dots, x_k$. The Dickson polynomial

$$f_V(X) \;=\; \prod_{v \in V} (X - v) \;=\; \sum_{i=0}^{k} c_{k,i}\, X^{p^i}, \qquad c_{k,k} = 1,$$

is additive in X. Its coefficients $c_{k,i}$ are the (signed) Dickson invariants. In the classical notation $c_{k,i} = (-1)^{k-i} Q_{k,i}$, and $|c_{k,i}| = (p^k - p^i)|x_1|$. Dickson's theorem says $D_k = H^{\mathrm{GL}_k} = \mathbb{F}_p[c_{k,0}, \dots, c_{k,k-1}]$ [1]. Two facts about additive polynomials will be used repeatedly:

(D1) An additive polynomial with $\mathbb{F}_p$-coefficients is $\mathbb{F}_p$-linear. So it vanishes on V as soon as it vanishes on $x_1, \dots, x_k$.

(D2) Over a domain, a monic polynomial of degree p^k that vanishes on the p^k distinct elements of V equals f_V.

Since Λ_t is a ring homomorphism,

$$\Lambda_t(f_V)(X) = \prod_{v \in V} \big(X - T(v)\big) = f_{T(V)}(X). \tag{2}$$

2.3. **The Mùi extension.** For odd p, Mùi [8] computed the full ring of invariants $H^*(BV_k)^{\mathrm{GL}_k}$. It is generated over D_k by exterior-type classes $R_{k,i}$ $(0 \le i < k)$ of degree $2(p^k-1)-2p^i+1$, with $R_{k,i}^2 = 0$. These are built from Moore-type determinants with one row of y's; see also [6]. The `DicksonAlgebra` class with `generic=True` adds odd generators τ_i of degree $1-2p^i$ to D_k. Their degrees are those of $R_{k,i}/Q_{k,0}$. The code treats them as images of Milnor's τ_i; see Section 6. This note concentrates on the polynomial part D_k.

3. Peterson elements

Let $S(z) = \sum_{j\ge 0} z^{p^j}$. This is the compositional inverse of the additive polynomial $z \mapsto z - z^p$, i.e.

$$S(z) - S(z)^p = z. \tag{3}$$

Definition 3.1. Let $e = x_1 \cdots x_k$. For $n \ge 0$ the *Peterson element* $\omega_n \in H$ is

$$\omega_n = e \cdot \chi(P^n)(e^{-1}).$$

Here e^{-1} lives in the localisation $H[e^{-1}]$, on which $\mathcal{A}$ acts by the Cartan formula. The *normalised* Peterson element is $\widetilde{\omega}_n = (-1)^n \omega_n$. It has degree $(p-1)n|x_1|$. (Code: `PetersonPolynomials.omega`, with `_omega_safe` implementing the definition.)

For $p = 2$ these are the elements studied by Hung and Peterson [5, §2].

Proposition 3.2. (1) *In the completion of H,*

$$\sum_{n\ge 0} \widetilde{\omega}_n \;=\; \prod_{i=1}^{k} \frac{x_i}{S(x_i)} \;=\; \prod_{i=1}^{k} (1 - S(x_i)^{p-1}).$$

(2) *Call $a \ge 0$* admissible *if $a = 0$ or the base-p digit sum of a is exactly $p-1$. For admissible $a = \sum_j \varepsilon_j p^j$, put $\varepsilon(a)! = \prod_j \varepsilon_j!$. Then*

$$\widetilde{\omega}_n \;=\; \sum_{\substack{a_1+\cdots+a_k=(p-1)n \\ a_i \ \text{admissible}}} \prod_{i=1}^{k} \frac{1}{\varepsilon(a_i)!}\, x_1^{a_1} \cdots x_k^{a_k}.$$

For $p = 2$ the admissible numbers are 0 and the powers of 2, and all coefficients are 1. This recovers the Hung–Peterson formula.

Proof. (1) Apply Λ'_t with $t = (-1, 0, 0, \dots)$. Then $T(x) = x - x^p$, so $\Lambda'_t(x) = T^{-1}(x) = S(x)$ by (3). Hence $\sum_n (-1)^n \chi(P^n)(e^{-1}) = \Lambda'_t(e^{-1}) = \prod_i S(x_i)^{-1}$. Multiply by e and take the homogeneous component of degree $(p-1)n|x_1|$. The second equality is $z/S(z) = (S(z) - S(z)^p)/S(z) = 1 - S(z)^{p-1}$.

(2) Expand $S(z)^{p-1} = \sum \binom{p-1}{\varepsilon_0,\varepsilon_1,\dots} z^{\sum \varepsilon_j p^j}$ over $\sum \varepsilon_j = p-1$. Different ε give different exponents, since there are no carries. By Wilson's theorem $-\binom{p-1}{\varepsilon} = -(p-1)!/\varepsilon! = 1/\varepsilon!$ in $\mathbb{F}_p$. □

Example 3.3. For $p = 3$, $k = 2$ we get $\widetilde{\omega}_1 = -(x_1^2 + x_2^2)$ and $\widetilde{\omega}_3 = -(x_1^6 + x_1^4 x_2^2 + x_1^2 x_2^4 + x_2^6) = c_{2,1}$. Also $\widetilde{\omega}_4 = x_1^6 x_2^2 + x_1^4 x_2^4 + x_1^2 x_2^6 = c_{2,0}$. The element $\widetilde{\omega}_1$ is not invariant. In general $\widetilde{\omega}_n$ need not be GL_k-invariant. For $p = 2$, $k = 3$, the non-invariant ones with $n \le 40$ are $n = 1, 2, 3, 5, 9, 17, 33$. For $p = 3$, $k = 2$ they are $n = 1, 2, 5, 14$.

Theorem 3.4 (Dickson invariants are Peterson elements)**.** *For $0 \le i \le k$,*

$$c_{k,i} = \widetilde{\omega}_{(p^k - p^i)/(p-1)}.$$

Proof. Let $F(X) = \sum_{i=0}^{k} \widetilde{\omega}_{(p^k-p^i)/(p-1)} X^{p^i}$. It is additive with $\mathbb{F}_p$-coefficients and monic of degree p^k, because $\widetilde{\omega}_0 = 1$. By (D1), (D2) and symmetry it suffices to show $F(x_1) = 0$. Introduce a grading variable w and put $\widetilde{\omega}(w) = \sum_n \widetilde{\omega}_n w^{(p-1)n} = \prod_l (1 - S(x_l w)^{p-1})$. The coefficient of w^{p^k} in $\widetilde{\omega}(w) S(x_1 w) = \sum_{i \geq 0} \widetilde{\omega}(w) (x_1 w)^{p^i}$ is $\sum_{i \leq k} \widetilde{\omega}_{(p^k-p^i)/(p-1)} x_1^{p^i} = F(x_1)$. By (3),

$$\widetilde{\omega}(w)\, S(x_1 w) = \bigl(S(x_1 w) - S(x_1 w)^p\bigr) \prod_{l \geq 2} (1 - S(x_l w)^{p-1}) = x_1 w \prod_{l=2}^{k} (1 - S(x_l w)^{p-1}).$$

Its w^{p^k}-coefficient is x_1 times a sum of monomials $x_2^{a_2} \cdots x_k^{a_k}$ with admissible a_l and $\sum a_l = p^k - 1$. The digit sum is subadditive, so the digit sum of $\sum a_l$ is at most $(k-1)(p-1)$. But $p^k - 1$ has digit sum $k(p-1)$. So there are no such monomials, and $F(x_1) = 0$. □

Remark 3.5. For $p = 2$, Theorem 3.4 is the formula of [5, §2]. In the code, generator number j of `DicksonAlgebra(p,k)` (counted from 1, in the order `d{deg}` is listed) is $c_{k,k-j} = \widetilde{\omega}_{n_j}$ with

$$n_j = p^{k-j} \frac{p^j - 1}{p - 1} = \frac{p^k - p^{k-j}}{p - 1}.$$

This is computed in `_left_steenrod_coaction_milnor_gen`. For example `d4,d6,d7` for $p = 2$, $k = 3$ are $\widetilde{\omega}_4, \widetilde{\omega}_6, \widetilde{\omega}_7$, and `d12,d16` for $p = 3$, $k = 2$ are $\widetilde{\omega}_3 = c_{2,1} = -Q_{2,1}$ and $\widetilde{\omega}_4 = c_{2,0} = Q_{2,0}$.

4. The algorithm

4.1. **Decomposing Peterson elements.** Let D_k^{perf} be the perfect closure of D_k, i.e. D_k with all p-power roots of its elements adjoined. Put

$$q_j = c_{k,k-j}^{\,p^{-(k-j)}} \in D_k^{\mathrm{perf}} \quad (0 \leq j \leq k), \qquad \varphi(u) = \sum_{j=0}^{k} q_j\, u^{p^j},$$

so $q_0 = 1$ and $q_k = c_{k,0}$. Since $\varphi(u) = u + O(u^p)$ is additive, it has an additive compositional inverse $\psi(w) = w + O(w^p)$.

Definition 4.1. For $n \geq 0$ and $m = (p-1)n$ let

$$\Omega_n = [u^m] \Bigl(\frac{u}{\varphi(u)}\Bigr)^m = \sum_R \binom{-m}{|R|} \binom{|R|}{r_1, r_2, \ldots, r_k} \prod_{j=1}^{k} q_j^{r_j} \ \in D_k^{\mathrm{perf}}, \tag{4}$$

where the sum runs over $R = (r_1, \ldots, r_k)$ with $\sum_j r_j (p^j - 1) = m$.

This is exactly what `PetersonPolynomials.decompose` and `DicksonAlgebra.peterson` compute. The Milnor basis enumerates the exponent vectors R. The term q^R is the Dickson monomial $\prod_j c_{k,k-j}^{\,r_j/p^{k-j}}$. When a term with nonzero coefficient has $p^{k-j} \nmid r_j$, the element lies in $D_k^{\mathrm{perf}} \setminus D_k$ and the code raises `ValueError`. Terms with $r_j \neq 0$ for some $j > k$ are dropped, which is consistent with $q_j = 0$ for $j > k$.

Proposition 4.2 (Lagrange inversion)**.** $\displaystyle\sum_{n \geq 0} \Omega_n\, w^{(p-1)n} = \frac{w}{\psi(w)}$.

Proof. Lagrange inversion in residue form holds over any commutative ring [2]. It says $[w^m]\, H(\psi(w)) = [u^m]\, H(u) \varphi'(u) (u/\varphi(u))^{m+1}$. Here $\varphi' = 1$ because φ is additive. Take $H(u) = \varphi(u)/u$. Also, $u/\varphi(u)$ is a series in u^{p-1}. □

Theorem 4.3. *$\Omega_{n_j} = c_{k,k-j}$ for $0 \leq j \leq k$. In particular the generators satisfy $\Omega_{n_j} = \widetilde{\omega}_{n_j}$.*

Proof. Put $\Omega(w) = w/\psi(w)$. As in the proof of Theorem 3.4, $\sum_j \Omega_{n_j} X^{p^{k-j}} = \left[w^{p^k}\right]\Omega(w)S(Xw)$. With $H(u) = \frac{\varphi(u)}{u}S(X\varphi(u))$, Lagrange inversion turns this into

$$\left[u^{p^k}\right]\, S(X\varphi(u))\Big(\frac{u}{\varphi(u)}\Big)^{p^k} = \left[u^{p^k}\right]\, S(X\varphi(u))\cdot\big(1+\textstyle\sum_{j\ge1} q_j^{p^k}\, u^{p^k(p^j-1)}\big)^{-1}.$$

The second factor is 1 plus terms of u-degree $\ge p^k$. The first has no constant term. So only the constant 1 contributes, and we get $\left[u^{p^k}\right]\sum_a X^{p^a}\varphi(u)^{p^a} = \sum_{a+j=k} q_j^{p^a} X^{p^a} = f_V(X)$. □

Conjecture 4.4. *If $\widetilde{\omega}_n$ is GL_k-invariant, then $\widetilde{\omega}_n = \Omega_n$.*

By Theorem 4.3 the conjecture holds at the generator degrees $n = n_j$. It has been verified in all cases listed in Table 1. The algorithm itself only needs the elements Ω_n, so it does not depend on Conjecture 4.4. The conjecture only matters for relating the code's `peterson(n)` to the polynomial $\widetilde{\omega}_n$ in other degrees.

4.2. **The action on Peterson elements.** The coefficient $a(R,n)$ of (1) is what `__action_exponents` computes for the ordinary action. The recursion `__action_exponents2` produces the multinomial $\binom{N(R)}{pr_1,p^2r_2,\dots}$ as an iterated product of binomial coefficients, and `binom_modp(p, psum+n*(p-1)-esum, psum)` is $\binom{(p-1)(n+d(R))}{N(R)}$. The operation Λ_t extends uniquely to D_k^{perf} (with values in $D_k^{\mathrm{perf}}[t^{1/p^\infty}]$). We let P^R act on D_k^{perf} by taking the coefficient of the integral monomial t^R.

Theorem 4.5 (Action formula)**.** *For all k, n and R,*

$$P^R\,\Omega_n = a(R,n)\,\Omega_{n+d(R)}.$$

This is proved in Section 5 (Theorem 5.11). The code uses a second formula for the conjugate action (`isconjugate=True`), proved in Theorem 5.14.

Theorem 4.6 (Conjugate action)**.** *For all k, n and R,*

$$\chi(P^R)\,\Omega_n = b(R,n)\,\Omega_{n+d(R)}, \qquad b(R,n) = \binom{-(p-1)n}{N(R)}\binom{N(R)}{pr_1,p^2r_2,\dots}.$$

The algorithm needs Theorem 4.5 only at the generator degrees $n = n_j$, where $\Omega_{n_j} = c_{k,k-j}$:

$$P^R c_{k,k-j} = a(R,n_j)\,\Omega_{n_j+d(R)}. \tag{5}$$

This case is proved in Theorem 5.9. If the coefficient is nonzero, then $\Omega_{n_j+d(R)} = P^R c_{k,k-j}/a(R,n_j)$ lies in D_k, so the call to `peterson` cannot fail.

Example 4.7. For $p=3$, $k=2$, $R=(0,1)$ and $n=n_1=3$ we have $d(R)=4$ and $N(R)=9$. The coefficient is $a(R,3)=\binom{14}{9}\equiv 1$, so $P(0,1)c_{2,1} = \Omega_7 = c_{2,1}c_{2,0}$. The code prints `P(0,1)*d12 = d12*d16`. For $R=(1)$ the coefficient is $\binom{8}{3}\equiv 2$, so $P^1c_{2,1} = 2c_{2,0}$, matching `P(1)*d12 = 2*d16`. In classical notation this reads $P^1Q_{2,1} = Q_{2,0}$.

5. Proofs and evidence

5.1. **A generating-function form.** Let $s_j = t_j^{1/p^j}$ and let

$$T^*(w) = \sum_{j\ge0}(t_j w)^{p^{-j}} = w + \sum_{j\ge1} s_j\, w^{p^{-j}}$$

be the adjoint of T. We work with formal sums of monomials $x^{\alpha}t^{\epsilon}w^{\beta}$ with $\epsilon \in \mathbb{Z}[1/p]^{J}_{\geq 0}$ and $\beta \in \mathbb{Z}[1/p]$, where each x-degree has only finitely many terms. These form a ring, and every expression below has only finitely many terms in each x-degree. Let π be the projection onto monomials with integral t-exponents. It is linear over elements with integral t-exponents.

Lemma 5.1. *Let $(\Omega_M)_{M\geq 0}$ be elements of degree $(p-1)M$ in a graded commutative $\mathbb{F}_p$-algebra, and $\Omega(w) = \sum_M \Omega_M w^{(p-1)M}$. Then*

$$\pi\Big[\sum_M \Omega_M\, T^*(w)^{(p-1)M}\Big] = \sum_{n\geq 0}\sum_R a(R,n)\, t^R\, \Omega_{n+d(R)}\, w^{(p-1)n}.$$

Proof. Write $T^*(w) = w(1+\sum_{j\geq 1}\vartheta_j)$ with $\vartheta_j = s_j w^{p^{-j}-1}$. Then $T^*(w)^{(p-1)M} = w^{(p-1)M}\sum_A \binom{(p-1)M}{|A|}\binom{|A|}{A}\vartheta^A$. The t_j-exponent of ϑ^A is A_j/p^j, which is integral iff $A_j = p^j r_j$. In that case $\vartheta^A = t^R w^{-(p-1)d(R)}$. Put $n = M - d(R)$. Terms with $n<0$ vanish, because then $(p-1)M < N(R) - |R| \leq N(R)$. □

On D_k^{perf} the total operation has fractional t-exponents, since $\Lambda_t(y^{1/p}) = \Lambda_t(y)^{1/p}$. For example, for $p=2$, $k=2$, $\Lambda_t(c_{2,1}^{1/2})$ contains $t_1^{1/2}c_{2,0}^{1/2}$. The operation P^R is the coefficient of the integral monomial t^R. So Theorem 4.5 says exactly that

$$\pi\big[\Lambda_t\,\Omega(w)\big] = \pi\big[\Omega(T^*w)\big], \qquad \Omega(w) = \frac{w}{\psi(w)}. \tag{6}$$

In degrees where $\Omega_n \in D_k$ (in particular for $k=1$, and at the generator degrees n_j), $\Lambda_t\Omega_n$ has integral exponents and the π on the left can be dropped. Informally: up to terms with fractional t-exponents, the total Steenrod operation acts on the Peterson series through the adjoint T^*. The mechanism is the identity

$$S\big(x\,T^*(w)\big) = S\big(T(x)\,w\big) + B(x,w), \qquad B(x,w) = \sum_{j\geq 1}\sum_{a<j} x^{p^a}(t_j w)^{p^{a-j}}, \tag{7}$$

obtained by splitting $\sum_{a,j} x^{p^a}(t_j w)^{p^{a-j}}$ according to whether $a \geq j$. Every monomial of B has t-exponent $p^{-c}\varepsilon_j$ with $c\geq 1$, where ε_j is the j-th unit vector.

Theorem 5.2. *Theorem 4.5 holds for $k=1$. Hence the algorithm of Section 4 is correct for D_1.*

Proof. For $k=1$, $\Omega_n = \widetilde{\omega}_n$, since both equal $(-1)^n\binom{-(p-1)n}{n}x^{(p-1)n}$. By Proposition 3.2, $\Omega(w) = 1 - S(xw)^{p-1}$. Therefore $\Lambda_t\Omega(w) = 1 - S(T(x)w)^{p-1}$ and $\Omega(T^*w) = 1 - S(xT^*w)^{p-1}$. By (7),

$$S(xT^*w)^e = \sum_{i=0}^{e}\binom{e}{i}S(T(x)w)^{e-i}B^i.$$

For $1\leq i\leq e\leq p-1$, every monomial of B^i has t-exponent vector equal to a sum of i vectors of the form $p^{-c}\varepsilon_j$ with $c\geq 1$. Each coordinate of such a sum is either 0 or lies in $(0,(p-1)/p]$, and at least one coordinate is nonzero, so the sum is not integral. Hence $\pi(B^i)=0$, and since $S(T(x)w)$ is integral, $\pi(S(xT^*w)^e) = S(T(x)w)^e$. So $\pi[\Omega(T^*w)] = \Lambda_t\Omega(w)$. Now compare coefficients using Lemma 5.1. □

5.2. **The general case at the generators.** For $k\geq 2$ the argument of Theorem 5.2, applied to $\prod_l(1 - S(x_l T^*w)^{p-1})$, fails. Fractional pieces coming from *different* factors can combine into integral monomials. These cross terms do not vanish in general; they are exactly why $(*)$ fails for non-invariant $\widetilde{\omega}_n$ (for example $p=2$, $k=2$, $n=1$, $R=(1)$). At the generators we can nevertheless prove the formula. The first step reduces it to a single identity.

Proposition 5.3. *Put*

$$F_t(X) = \Big[w^{p^k}\Big]\ \pi[\Omega(T^*w)]\, S(Xw) = \sum_{j=0}^{k}\Big(\sum_R a(R,n_j)\, t^R\, \Omega_{n_j+d(R)}\Big)X^{p^{k-j}}.$$

Then (5) *holds for all j and R if and only if $F_t(T(x_1)) = 0$.*

Proof. The two expressions for F_t agree by Lemma 5.1. For $j = 0$ we have $a(R,0) = \binom{N(R)-|R|}{N(R)} = 0$ unless $R = 0$, so F_t is additive and monic of degree p^k, with coefficients in $D_k^{\mathrm{perf}}[t]$. It is symmetric in the x_i. So if $F_t(T(x_1)) = 0$, then F_t vanishes on $T(V)$, and by (D2) $F_t = f_{T(V)} = \Lambda_t f_V$ by (2). Comparing coefficients gives (5). The converse is the same computation read backwards. □

The rest of this subsection proves $F_t(T(x_1)) = 0$. We need three tools: a factorisation of the Peterson series, a bound on its support, and an explicit formula for $F_t(T(X))$. Throughout, $\alpha(m)$ denotes the base-p digit sum of $m \geq 0$. Recall that $\alpha(a+b) \leq \alpha(a)+\alpha(b)$, with equality iff the addition has no carries. By Lucas' theorem, a multinomial coefficient $\binom{a_1+\cdots+a_m}{a_1,\ldots,a_m}$ is nonzero mod p iff the addition $a_1+\cdots+a_m$ has no carries. Write $g(z) = 1 - S(z)^{p-1} = z/S(z)$ and $\gamma_m = [z^m]\, g(z)$. By Proposition 3.2, $\gamma_m \neq 0$ only if $m = 0$ or $\alpha(m) = p-1$.

Factorisation. Let $0 \neq v \in V$, $f_v(X) = X^p - v^{p-1}X$ and $U = f_v(V)^{1/p}$, an $\mathbb{F}_p$-space of rank $k-1$. Put $\psi_v(w) = S(vw)/v$.

Lemma 5.4. $\Omega_V(w) = g(vw)\cdot \Omega_U(\psi_v(w))$.

Proof. By (D2), $f_V = f_{f_v(V)} \circ f_v$. Write additive series as twisted polynomials in the Frobenius F, with adjoint $(aF^i)^* = F^{-i}a$ and $(AB)^* = B^*A^*$. Then $\varphi_V = f_V^* F^k$, and

$$\varphi_V = f_v^*\, f_{f_v(V)}^*\, F^k = (f_v^* F)\circ (F^{-1}(f_{f_v(V)}^* F^{k-1})F) = \varphi_{\langle v\rangle}\circ\varphi_U,$$

because conjugating by F takes p-th roots of the coefficients. Hence $\psi_V = \psi_U \circ \psi_{\langle v\rangle}$ and $w/\psi_V(w) = \frac{w}{\psi_{\langle v\rangle}(w)}\cdot\frac{\psi_{\langle v\rangle}(w)}{\psi_U(\psi_{\langle v\rangle}(w))}$. For rank one, $\varphi_{\langle v\rangle}(u) = u - v^{p-1}u^p$, so $\psi_{\langle v\rangle} = \psi_v$ and $w/\psi_v(w) = vw/S(vw) = g(vw)$. □

Lemma 5.5 (Support)**.** *Let $\eta(w) = \sum_{a\geq 0} e_a w^{p^a}$ be any additive series. Then $\Big[w^E\Big]\ \Omega_V(\eta(w)) = 0$ unless $\alpha(E) \leq k(p-1)$.*

Proof. Induction on k; for $k = 0$, $\Omega_V = 1$. By Lemma 5.4, $\Omega_V(\eta(w)) = g(v\eta(w))\,\Omega_U(\eta'(w))$ with $\eta' = \psi_v\circ\eta$ additive. The first factor is $1 - S(v\eta(w))^{p-1}$, and $S(v\eta(w))$ is additive in w. So the first factor is supported in exponents that are 0 or a sum of $p-1$ powers of p, i.e. of digit sum $\leq p-1$. By induction the second factor is supported in digit sum $\leq (k-1)(p-1)$. Now use subadditivity of α. □

For $D_0 \geq 1$ put

$$F_t^{D_0}(X) = \Big[w^{D_0}\Big]\ \pi[\Omega(T^*w)]\, S(Xw),$$

so that $F_t = F_t^{p^k}$. This is an additive polynomial in X.

Lemma 5.6. *Fix $D_0 \geq 1$. For R put $D_R = D_0 + (p-1)d(R)$, and write $\Big[y^{pR}\Big]$ for the coefficient of $\prod_j y_j^{p^j r_j}$. Put*

$$b_R = \Big[y^{pR}\Big]\,(1+\textstyle\sum_j y_j)^{D_R}, \qquad \delta_{R,e} = \Big[y^{pR}\Big]\,\Big(\sum_{j>e} y_j^{p^e}\Big)(1+\textstyle\sum_j y_j)^{D_R - p^e}.$$

Then

$$F_t^{D_0}(T(X)) = \sum_R t^R \left[w^{D_R}\right]\, \Omega(w)\, B_R(Xw), \qquad B_R(z) = b_R\, S(z) - \sum_{e\geq 0} \delta_{R,e}\, z^{p^e}.$$

Proof. Since $S(T(X)w)$ has integral t-exponents, $F_t^{D_0}(T(X)) = \pi\left[w^{D_0}\right]\Omega(T^*w)\,S(T(X)w)$. By (7), $S(T(X)w) = S(XT^*w) - B(X,w)$. The first part is $\pi\left[w^{D_0}\right]H(T^*w)$ with $H(w) = \Omega(w)S(Xw)$. Expanding as in Lemma 5.1 gives $\sum_R t^R b_R\left[w^{D_R}\right]H(w)$. For the second part, take a monomial $X^{p^a}(t_j w)^{p^{a-j}}$ of B (so $a<j$) times the term $\Omega_M w^{(p-1)M}\binom{(p-1)M}{|A|}\binom{|A|}{A}\vartheta^A$ of $\Omega(T^*w)$. Its t-exponents are integral iff $A_i = p^i r_i$ for $i\neq j$ and $A_j = p^j r_j - p^a$ with $r_j\geq 1$. Then its w-exponent is $(p-1)M - (p-1)d(R) + p^a$, which equals D_0 iff $(p-1)M = D_R - p^a$. Summing the coefficients over $j>a$ gives $\delta_{R,a}$. So the second part is $-\sum_R t^R \sum_a \delta_{R,a} X^{p^a}\Omega_{(D_R-p^a)/(p-1)} = -\sum_R t^R\left[w^{D_R}\right]\Omega(w)\sum_a \delta_{R,a}(Xw)^{p^a}$. □

Lemma 5.7 (Digit sums)**.** *Let $D_0\geq 1$, R arbitrary, $e\geq 0$ and $D = D_R = D_0 + (p-1)d(R)$.*

1. *If $\left[y^{pR}\right](1+\sum_j y_j)^{D-1}\neq 0$, then $\alpha(D-1)\geq \alpha(D_0-1)$.*
2. *If $\delta_{R,e}\neq 0$, then $\alpha(D-p^e)\geq \alpha(D_0-1)+p-1$.*

Proof. Recall $D-N(R) = D_0-|R|$. (1) The coefficient is $\binom{D-1}{N(R)}\binom{N(R)}{pR}$. If it is nonzero, the parts $p^j r_j$ add without carries, so $\alpha(N(R)) = \sum_j \alpha(r_j)\geq \alpha(|R|)$. Moreover $D-1-N(R) = D_0-1-|R|\geq 0$ and $N(R)+(D_0-1-|R|)$ has no carries. Since $\alpha(D_0-1)\leq \alpha(D_0-1-|R|)+\alpha(|R|)$, we get $\alpha(D-1) = \alpha(N(R))+\alpha(D_0-1-|R|)\geq \alpha(D_0-1)$.

(2) Some summand $\binom{D-p^e}{N(R)-p^e}\binom{N(R)-p^e}{pR-p^e\varepsilon_j}$ with $j>e$ and $r_j\geq 1$ is nonzero. The parts $p^i r_i$ $(i\neq j)$ and $p^j r_j - p^e = p^e(p^{j-e}(r_j-1)+p^{j-e}-1)$ add without carries, so

$$\alpha(N(R)-p^e) = \sum_{i\neq j}\alpha(r_i) + \alpha(r_j-1) + (j-e)(p-1).$$

The binomial coefficient is nonzero, so $D-N(R) = D_0-|R|\geq 0$ (with $|R|\geq 1$) and $\alpha(D-p^e) = \alpha(N(R)-p^e)+\alpha(D_0-|R|)$. Moreover $\alpha(D_0-|R|)\geq \alpha(D_0-1)-\alpha(|R|-1)$ and $\alpha(|R|-1)\leq \sum_{i\neq j}\alpha(r_i)+\alpha(r_j-1)$. Hence

$$\alpha(D-p^e)\geq (j-e)(p-1)+\alpha(D_0-1)\geq \alpha(D_0-1)+p-1.$$ □

Proposition 5.8 (Vanishing)**.** *If $\alpha(D_0-1) > (k-1)(p-1)$, then $F_t^{D_0}(T(v)) = 0$ for all $v\in V$.*

Proof. By additivity it suffices to take $v\neq 0$. By Lemma 5.4, $\Omega_V(w) = g(vw)G(w)$ with $G(w) = \Omega_U(\psi_v(w))$. Since $g(z)S(z) = z$,

$$g(z)B_R(z) = \sum_E G_{R,E}\, z^E, \qquad G_{R,E} = b_R\,[E=1] - \sum_{e\geq 0}\delta_{R,e}\,\gamma_{E-p^e}.$$

By Lemma 5.6, the coefficient of t^R in $F_t^{D_0}(T(v))$ is $\sum_E G_{R,E}\, v^E\left[w^{D_R-E}\right]G(w)$. We show that every term vanishes. Suppose $\left[w^{D_R-E}\right]G\neq 0$. By Lemma 5.5 (rank $k-1$, $\eta=\psi_v$), $\alpha(D_R-E)\leq (k-1)(p-1)$.

- If $E=1$, then $G_{R,1} = b_R - \delta_{R,0} = \left[y^{pR}\right](1+\sum y_j)^{D_R-1}$. If this were nonzero, Lemma 5.7(1) would give $\alpha(D_R-1)\geq \alpha(D_0-1) > (k-1)(p-1)$. This is a contradiction.

- If $E > 1$, a nonzero summand $\delta_{R,e}\gamma_{E-p^e}$ has $\alpha(E-p^e) \le p-1$. Hence $\alpha(D_R - p^e) \le \alpha(D_R - E) + \alpha(E-p^e) \le k(p-1)$. But Lemma 5.7(2) gives $\alpha(D_R - p^e) \ge \alpha(D_0 - 1) + p - 1 > k(p-1)$.

So $G_{R,E} = 0$ whenever $\left[w^{D_R - E}\right] G \neq 0$, and $F_t^{D_0}(T(v)) = 0$. □

Theorem 5.9 (Action on the generators)**.** *Formula* (5) *holds for all k, j and R:*

$$P^R c_{k,k-j} = a(R, n_j)\, \Omega_{n_j + d(R)}.$$

Proof. Take $D_0 = p^k$, for which $\alpha(p^k - 1) = k(p-1) > (k-1)(p-1)$. Proposition 5.8 gives $F_t(T(x_1)) = 0$; now apply Proposition 5.3. □

Corollary 5.10. *The algorithm of Section 4 computes the correct $\mathcal{A}$-action on D_k for all p and k. Moreover, if $a(R, n_j) \neq 0$, then $\widetilde{\omega}_{n_j + d(R)} = \Omega_{n_j + d(R)}$. So Conjecture 4.4 holds in all degrees that the algorithm uses.*

Proof. The first claim follows from Theorem 5.9 and Section 6. For the second, run the same proof with the polynomial family $\widetilde{\omega}_n$ in place of Ω_n. Lemmas 5.1 and 5.6 hold for any family. The factorisation is $\widetilde{\omega}(w) = g(x_1 w) \prod_{l \ge 2} g(x_l w)$ (Proposition 3.2), and the second factor is supported in digit sum $\le (k-1)(p-1)$. So $P^R c_{k,k-j} = a(R, n_j)\, \widetilde{\omega}_{n_j + d(R)}$ as well. □

For single powers P^i, Theorem 5.9 recovers the known formulas of Hung [3] ($p = 2$) and Hung–Minh [4] (odd p; see also [10, Thm. 3.4]). We compared these with the code's output for all s and i for $p = 2$, $k = 3, 4$; $p = 3$, $k = 2, 3$; and $p = 5, 7$, $k = 2$, using $c_{k,s} = (-1)^{k-s} Q_{k,s}$. All 298 values agree.

5.3. **All degrees.**

Theorem 5.11. *Theorem 4.5 holds: $P^R \Omega_n = a(R, n)\, \Omega_{n + d(R)}$ for all k, n and R.*

Proof. Put $G = \pi[\Omega_V(T^* w)]$ and $G' = \pi[\Lambda_t \Omega_V(w)] = \pi[\Omega_{T(V)}(w)]$. By Lemma 5.1 we must show $G = G'$. Let $\Delta = G' - G = \sum_n \Delta_n w^{(p-1)n}$. Then $\Delta_0 = 0$, and every Δ_n has integral t-exponents.

Step 1: a family of linear conditions. Let $\alpha(D_0 - 1) > (k-1)(p-1)$ and $u \in T(V)$. By Proposition 5.8, $\left[w^{D_0}\right] G(w) S(uw) = F_t^{D_0}(u) = 0$. Lemmas 5.4 and 5.5 hold for the rank-k space $T(V)$ as well. So $\Omega_{T(V)}(w) S(uw) = uw\, \Omega_{U'}(\psi_u(w))$ is supported in exponents E with $\alpha(E-1) \le (k-1)(p-1)$. Since $S(uw)$ is integral, the same holds for $G'(w) S(uw)$. Hence the additive polynomial

$$L_{D_0}(X) = \left[w^{D_0}\right] \Delta(w) S(Xw) = \sum_{e \ge 0} \Delta_{(D_0 - p^e)/(p-1)} X^{p^e}$$

vanishes on $T(V)$. Divide L_{D_0} by the monic additive polynomial $f_{T(V)}$: $L_{D_0} = M \circ f_{T(V)} + \mathrm{Rem}$, where Rem has degree $< p^k$. The usual division algorithm works for additive polynomials over any ring of characteristic p. Since Rem vanishes on the p^k elements of $T(V)$, $\mathrm{Rem} = 0$.

Step 2: a recursion. Let $C = \{n \ge 1 : p^{k-1} \mid (p-1)n\}$. For $n \in C$ put $D_0 = (p-1)n + p^{k-1}$. The low $k-1$ digits of $(p-1)n$ vanish, so $\alpha(D_0 - 1) = \alpha((p-1)n) + (k-1)(p-1) > (k-1)(p-1)$. In L_{D_0}:

- the term $e = k-1$ is $\Delta_n X^{p^{k-1}}$;
- the terms $e < k-1$ are X^{p^e} with $e < k-1$, and are already reduced;
- the terms $e \ge k$ have indices $n_e = n - (p^e - p^{k-1})/(p-1) < n$. Here $(p-1)n_e \equiv 0 \bmod p^{k-1}$, so $n_e \in C \cup \{0\}$, or the term is absent.

Reducing X^{p^e} ($e \geq k$) with $X^{p^k} \equiv -\sum_{i<k} c_i(T(V))X^{p^i}$, the coefficient of $X^{p^{k-1}}$ in Rem is $\Delta_n + \sum_{e\geq k} \rho_e \Delta_{n_e}$ for certain ρ_e. It vanishes by Step 1. By induction on $n \in C$, starting from $\Delta_0 = 0$, we get $\Delta_n = 0$ for all $n \in C$.

Step 3: integral degrees. Let $N = p^k m$ with $m \geq 1$; then $N \in C$. Moreover $\Omega_N = \Omega_m^{p^k}$: raise $(u/\varphi)^{(p-1)m}$ to the p^k-th power in (4). This lies in D_k, because $q_j^{p^k} \in D_k$. So $\Lambda_t\Omega_N$ has integral t-exponents, and Step 2 gives

$$\Lambda_t\Omega_N = \sum_S t^S\, a(S,N)\, \Omega_{N+d(S)}.$$

Step 4: descent. For arbitrary m, $\Lambda_t\Omega_m = (\Lambda_t\Omega_{p^k m})^{p^{-k}}$. The coefficient of the integral monomial t^R comes from $S = p^k R$. It is $\big(a(p^kR, p^km)\,\Omega_{p^k(m+d(R))}\big)^{p^{-k}} = a(R,m)\,\Omega_{m+d(R)}$, by Lucas' theorem and $\Omega_{p^k j} = \Omega_j^{p^k}$. □

For $k = 1$, C consists of all $n \geq 1$, and Step 2 is Theorem 5.2 again.

5.4. **The conjugate action.** On D_k^{perf} we let $\chi(P^R)$ act by the integral coefficients of $\Lambda_t' = \Lambda_t^{-1}$, as for P^R.

The proof reduces the conjugate formula to the ordinary one, using the antipode identity and a vanishing statement for binomial coefficients.

Proposition 5.12. *Suppose that, for all $n \geq 0$ and $R \neq 0$,*

$$\sum_{R'+R''=R} b(R',\, n + d(R''))\; a(R'',\, n) = 0. \tag{8}$$

Then Theorem 4.6 holds.

Proof. By the same descent as in Step 4, it suffices to treat N with $\Omega_N \in D_k$. The element $\chi(P^R)$ is a linear combination of Milnor basis elements P^S with $d(S) = d(R)$. So by Theorem 5.11, $\chi(P^R)\Omega_N$ is a scalar multiple of $\Omega_{N+d(R)}$. We show by induction on $|R|$ that the scalar is $b(R,N)$, for all such N. Apply $\sum_{R'+R''=R} \chi(P^{R'})P^{R''} = 0$ (the antipode identity, $R \neq 0$) to Ω_N. If $a(R'',N) \neq 0$, then $\Omega_{N+d(R'')} = P^{R''}\Omega_N / a(R'',N)$ lies in D_k, so for $R'' \neq 0$ the induction hypothesis applies to it. Thus

$$\chi(P^R)\Omega_N = -\sum_{R''\neq 0} b(R', N + d(R''))\, a(R'', N)\, \Omega_{N+d(R)} = b(R,N)\,\Omega_{N+d(R)}$$

by (8), since $a(0,\cdot) = 1$. □

To prove (8) we need a projection on power series. Let $y_1, \ldots, y_J$ be variables, $Y = 1 + \sum_i y_i$, and fix levels $\lambda = (\lambda_1, \ldots, \lambda_J) \in \mathbb{Z}_{\geq 0}^J$. Let Π_λ keep the monomials y^E with $p^{\lambda_i} \mid E_i$ for all i, and put $\|E\|_\lambda = \sum_i E_i/p^{\lambda_i}$ for such E. For $m \in \mathbb{Z}$, Y^m is a power series, a polynomial if $m \geq 0$.

Lemma 5.13. *Let $E \neq 0$ with $p^{\lambda_i} \mid E_i$, and let $\mu, \nu \in \mathbb{Z}$ with $\mu + \nu = |E| - \delta$, where $1 \leq \delta \leq \|E\|_\lambda$. Then*

$$\big[y^E\big]\; \Pi_\lambda(Y^\mu)\, \Pi_\lambda(Y^\nu) = 0.$$

Proof. Write $m = \sum_a m_a p^a$ in base p (p-adically if $m < 0$). By Lucas, $Y^m = \prod_a (1 + \sum_i y_i^{p^a})^{m_a}$. In this product the exponent of y_i is $\sum_a p^a k_{a,i}$ with digits $k_{a,i} \leq m_a < p$, so it is divisible by

p^{λ_i} iff $k_{a,i} = 0$ for $a < \lambda_i$. Hence

$$\Pi_\lambda(Y^m) = \prod_{a\ge 0}\Big(1 + \sum_{i:\lambda_i\le a} y_i^{p^a}\Big)^{m_a} = \Big(1+\sum_{i\in Z} y_i\Big)^{m_0}\cdot \Pi_{\lambda^-}(Y^{\lfloor m/p\rfloor})^p,$$

where $Z = \{i : \lambda_i = 0\}$, $\lambda_i^- = \max(\lambda_i - 1, 0)$ and $\lfloor m/p\rfloor = (m - m_0)/p$. We use induction on $\sum_{i\notin Z,\, E_i\neq 0}\lambda_i$.

If $E_i = 0$ for all $i \notin Z$, set those $y_i = 0$. Then $\Pi_\lambda(Y^m)$ becomes $(1+\sum_{i\in Z} y_i)^m$, and the coefficient is $\binom{|E|-\delta}{|E|}\binom{|E|}{E} = 0$, since $0 \le |E| - \delta < |E|$.

Otherwise let $s = \mu_0 + \nu_0 \le 2p-2$, $\mu' = \lfloor \mu/p\rfloor$, $\nu' = \lfloor \nu/p\rfloor$. Then the coefficient is a sum, over splittings $E_i = e_i + pe_i'$ ($i \in Z$), of $[y^e]\,(1+\sum_{i\in Z} y_i)^s$ times $\left[y^{E''}\right]\Pi_{\lambda^-}(Y^{\mu'})\Pi_{\lambda^-}(Y^{\nu'})$. Here $E_i'' = E_i/p$ for $i \notin Z$ and $E_i'' = e_i'$ for $i \in Z$. Only splittings with $|e| \le s$ contribute. We check the hypotheses for $(\lambda^-, E'', \mu', \nu')$. We have $|E| = p|E''| + |e|$, so $\mu'+\nu' = (\mu+\nu-s)/p = |E''| - \delta''$ with $\delta'' = (\delta + s - |e|)/p$, an integer. Since $\delta \ge 1$ and $|e| \le s$, $\delta'' \ge 1$. Let $c = \sum_{i\notin Z} E_i/p^{\lambda_i} \ge 1$, so that $\|E\|_\lambda = c + |e| + p|e'|$ and $\|E''\|_{\lambda^-} = c + |e'|$. Then $\delta'' \le (c+s)/p + |e'| < c + 1 + |e'|$, using $s \le 2p-2 < (p-1)c + p$. So $\delta'' \le \|E''\|_{\lambda^-}$. If $E'' = 0$, then $|E| = |e| \le s$ and $c = 0$, which is excluded. So the induction hypothesis applies, and every term vanishes. □

Theorem 5.14. *Identity* (8) *holds. Hence Theorem 4.6 holds.*

Proof. Let $\rho_a(t)$ and $\rho_b(t)$ be the operators on formal symbols $(E_n)_{n\ge 0}$ given by $E_n \mapsto \sum_R t^R a(R,n) E_{n+d(R)}$ and $E_n \mapsto \sum_R t^R b(R,n) E_{n+d(R)}$. Identity (8) says $\rho_b\rho_a = 1$. Both operators are the identity plus terms of positive t-degree, so this is equivalent to $\rho_a\rho_b = 1$, i.e. to

$$\sum_{R_1+R_2=R} b(R_1,n)\, a(R_2, n+d(R_1)) = 0 \qquad (R \neq 0).$$

Now $b(R_1,n) = \left[y^{pR_1}\right] Y^{-(p-1)n}$, where $y^{pR} = \prod_j y_j^{p^j r_j}$. Similarly $a(R_2, n+d(R_1)) = \left[y^{pR_2}\right] Y^{(p-1)n + N(R) - |R|}$, because $(p-1)(n + d(R_1) + d(R_2)) = (p-1)n + N(R) - |R|$. So the sum is $\left[y^{pR}\right]\Pi_\lambda(Y^\mu)\Pi_\lambda(Y^\nu)$ with $\lambda_j = j$, $\mu = -(p-1)n$, $\nu = (p-1)n + N(R) - |R|$. Here $E = pR$ has $|E| = N(R)$ and $\|E\|_\lambda = |R|$, and $\mu + \nu = |E| - |R|$. Lemma 5.13 with $\delta = |R|$ shows that it vanishes. Now apply Proposition 5.12. □

5.5. **Computer verification.** All checks below compute the action directly on polynomials in $\mathbb{F}_p[x_1, \dots, x_k]$, by substituting $x_i \mapsto T(x_i)$ (or $x_i \mapsto T^{-1}(x_i)$ for χ) and extracting coefficients. They are independent of yacop's implementation. Every check that was run succeeded.

The module doctests also check associativity $(ab)m = a(bm)$ of the action the code defines (`_manual_test_left_action`).

6. The action on the whole algebra

The code computes $P^R(m)$ for a monomial m through a *coaction*: a list of triples (y, Q, R'), standing for $y \otimes \xi^{R'}$ (and exterior τ^Q). Then $P^R(m)$ is the sum of the y with $R' = R$ (`_act_from_coact` in `algebra_base.py`). The ingredients are:

(1) *Generators.* For $c_{k,k-j}$ the coaction is $\sum_R a(R,n_j)\,\Omega_{n_j+d(R)} \otimes \xi^R$, truncated to the R that are dominated by the requested operation.

(2) *Powers.* The function `_factor_key` writes an exponent in base p, $g^e = \prod_u (g^{p^u})^{\varepsilon_u}$ with $0 \le \varepsilon_u < p$. The coaction of g^{p^u} is that of g with everything raised to the p^u-th power, since $P^R(y^{p^u}) = (P^{R/p^u} y)^{p^u}$, or 0 if $p^u \nmid R$. For $\varepsilon_u > 1$ a multinomial expansion is used (`_multinomials`).

statement	range	cases
$(*)$ for $\widetilde{\omega}_n$, $k=1$ (exact formula)	$p=3,5,7$, $n<60$	1350
$(*)$, invariant $\widetilde{\omega}_n$	$p=2$: $k=2,3$; $p=3$: $k\le 3$; $p=5,7$: $k=2$	682
conjugate formula, invariant $\widetilde{\omega}_n$	$p=2$: $k=2,3$; $p=3$: $k\le 3$; $p=5$: $k\le 2$	321
Conjecture 4.4	$p=2$: $k=3,4$, $n\le 40$; $p=3$: $k=2$, $n\le 40$; $k=3$, $n\le 30$; $p=5$: $n\le 40$; $p=7$: $n\le 30$ $(k=2)$	143
$(*)$ for Ω_n, all n (in D_k^{perf})	$p=2$: $k=2,3$; $p=3$: $k=2$	134
$(*)$ for $P^{\Delta_j}=P(0,\dots,0,1)$, invariant $\widetilde{\omega}_n$	$p=2$: $k=3$ $(j=2,3)$, $k=4$ $(j=2)$; $p=3$: $k=3$, $j=2$; $p=5$: $k=2$, $j=2$	75
$G_{R,E}=0$ when $\alpha(D_R-E)\le(k-1)(p-1)$	$p=2,3,5$; $k\le 4$; $\ell(R)\le 3$, $r_j\le p+1$	$\approx 10^6$
same, general D_0 (Prop. 5.8)	$p=2,3,5$; $k\le 3$; $\ell(R)\le 2$; $D_0<p^{k+1}+p$	$1.3\cdot 10^6$
identity (8)	$p=2,3$: $\ell(R)\le 3$; $p=5$: $\ell(R)\le 2$; $n<25$	7175
Lemma 5.13	$p=2,3$; $J\le 3$; $\lambda_i\le 2$; $\lvert\mu\rvert\le 25$	240516

TABLE 1. Computer checks of the action formula. "Cases" counts pairs (n,R) (or values of n for the decomposition), with R ranging over small Milnor indices.

(3) *Products.* Coactions of factors are multiplied (`_coaction_helper`). This is the Cartan formula $P^R(yz)=\sum_{R'+R''=R}P^{R'}(y)P^{R''}(z)$, i.e. multiplicativity of Λ_t.

(4) *Negative powers.* In the localised algebra (the elements `_zetapower`, `_xipower` involve $c_{k,0}^{-1}$), a negative exponent $-e$ is written as $-p^u+(p^u-e)$, with p^u larger than every entry of R. Then $c_{k,0}^{-p^u}$ is invariant under all relevant $P^{R'}$.

(5) *Mùi generators.* For τ_n the code uses Milnor's coproduct $\Delta\tau_n=\tau_n\otimes 1+\sum_i\xi_{n-i}^{p^i}\otimes\tau_i$. It reads ξ_j as the element `_xipower(j)` of the localised Dickson algebra, defined by $\xi_j=-\sum_{i<j}\xi_i\zeta_{j-i}^{p^i}$ with $\zeta_i=c_{k,i}/c_{k,0}$. We have not verified this part independently. The source marks the corresponding conjugate formula with a `FIXME`.

Because D_k is a polynomial algebra, (1)–(3) determine the action completely. Together with Theorem 5.9 this proves that the D_k part of the code is correct for the ordinary action. The conjugate action (`left_steenrod_action_milnor_conj`) uses Theorem 4.6 on the Dickson generators, so its D_k part is correct too. The code also offers an independent route, `admissible_action`, through admissible matrices, and restriction maps $D_k\to D_{k'}$ for $k'<k$; these are not discussed here.

7. The dual Dickson algebras

The class `DicksonDualSteenrodAlgebra(`p`,`k`)` implements

$$\mathcal{A}_*(k)=\mathbb{F}_p[\xi_1,\dots,\xi_k]\otimes E(\tau_0,\dots,\tau_{k-1})\subset\mathcal{A}_*,$$

with the generators placed in negative degrees ($|\xi_j|=-2(p^j-1)$, $|\tau_j|=1-2p^j$). The code prints the ξ_j as `zeta[j]`. Milnor's coproduct $\Delta\xi_n=\sum_i\xi_{n-i}^{p^i}\otimes\xi_i$ only involves indices $\le n$, and similarly for τ_n. So $\mathcal{A}_*(k)$ is a sub-bialgebra of $\mathcal{A}_*$. It is an $\mathcal{A}$-module algebra via $\theta\cdot y=\sum y_{(1)}\langle\theta,y_{(2)}\rangle$ (`_coproduct_gen` and `_left_steenrod_coaction_milnor_gen`).

It is related to the Dickson algebras in two ways.

(a) The localised Dickson algebra. The algebra map $\xi_i\mapsto c_{k,i}/c_{k,0}$ (`DicksonAlgebra._zetapower`) sends $\mathcal{A}_*(k)$'s polynomial part into $D_k[c_{k,0}^{-1}]$ and preserves degrees. The code uses its conjugate (`_xipower`) for the Mùi generators, as explained in Section 6.

(b) Peterson elements are images of a canonical element of $\mathcal{A}_*$. Let $\xi(w) = \sum_j \xi_j w^{p^j}$ and $\zeta(w) = \sum_j \chi(\xi_j) w^{p^j}$. These are compositional inverses of each other, because $\sum_i \chi(\xi_i)\xi_{n-i}^{p^i} = 0$ for $n \geq 1$. Let $\mu\colon \mathcal{A}_* \to D_k^{\mathrm{perf}}$ be the algebra map with $\chi(\xi_j) \mapsto q_j$ for $j \leq k$ and $\chi(\xi_j) \mapsto 0$ for $j > k$. Then $\mu(\zeta) = \varphi$, $\mu(\xi) = \psi$, and by Proposition 4.2

$$\sum_n \Omega_n\, w^{(p-1)n} = \mu\Big(\frac{w}{\xi(w)}\Big) = \mu\Big(\sum_R (-1)^{|R|} \binom{|R|}{R} \xi^R\, w^{(p-1)d(R)}\Big).$$

The roots of φ form an $\mathbb{F}_p$-subspace Y of dimension k, in degree $-|x_1|$, of an algebraic closure. Since φ has linear coefficient 1 and top coefficient $c_{k,0}$, we have $\varphi = f_Y/c_0(Y)$ and $q_j = c_j(Y)/c_0(Y)$. So, up to the conjugation χ, μ is the map of (a) for the "dual" space Y in place of V. In this sense the Peterson elements of V are the images of the canonical elements $\sum_{d(R)=n}(-1)^{|R|}\binom{|R|}{R}\xi^R \in \mathcal{A}_*(k)$ under the Dickson map of the dual space. Equation (6), with the adjoint T^*, suggests that this duality is the natural setting for these formulas.

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